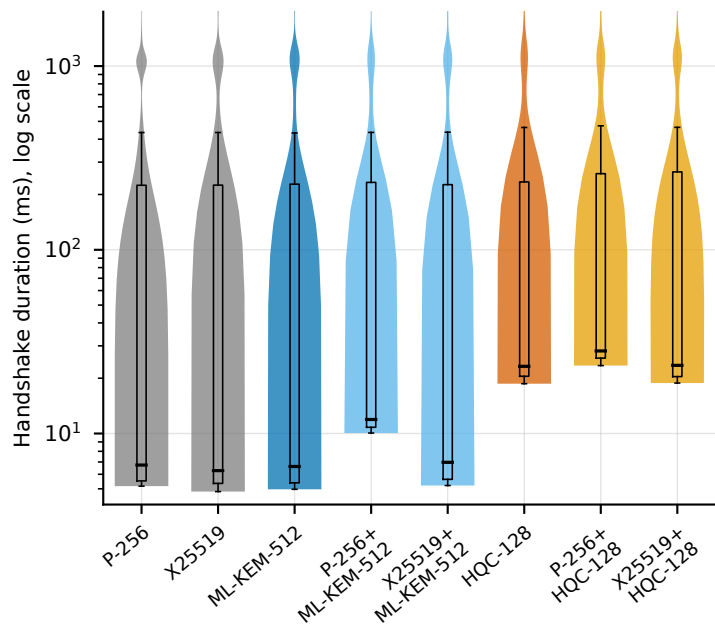
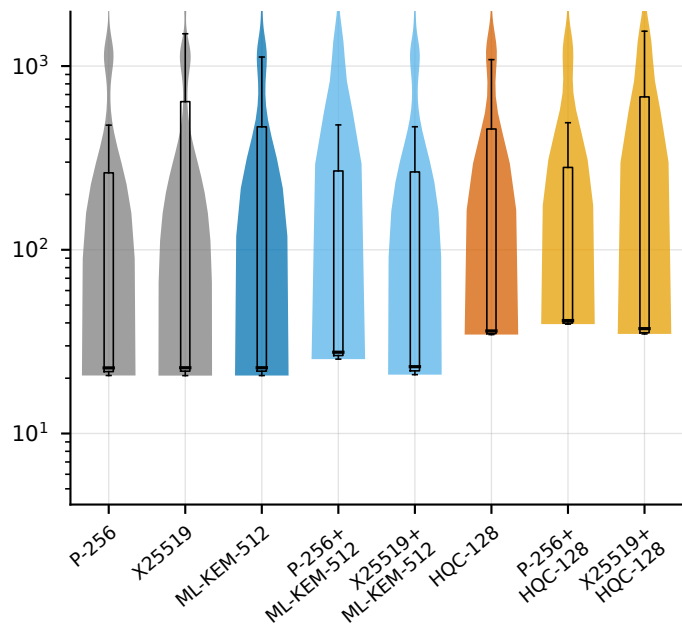


(a)

ed25519 (classical)

**(b)**

mldsa44 (post-quantum)

**(c)**
 Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	ed25519 Mean (ms)	ed25519 CV (%)	ed25519 Max (ms)	mldsa44 Mean (ms)	mldsa44 CV (%)	mldsa44 Max (ms)
P-256	244.5	176.6	2.10k	331.3	207.1	6.84k
X25519	306.8	204.6	5.16k	384.0	170.5	4.56k
ML-KEM-512	292.1	196.7	4.40k	392.9	204.8	9.72k
P-256+ML-KEM-512	283.4	229.7	7.63k	414.9	327.6	26.51k
X25519+ML-KEM-512	289.9	219.8	7.19k	388.9	199.3	7.02k
HQC-128	335.0	219.5	7.62k	410.9	216.5	12.92k
P-256+HQC-128	350.1	200.9	7.27k	442.4	225.7	13.39k
X25519+HQC-128	340.6	176.2	3.37k	505.4	269.2	26.21k